A Error Analysis of Horner-Form lg with Iterative Square-Root Range Reduction

We normalize $X = 2^e V$ with $V \in [1, 2)$, and apply M rounds of square-root range reduction. Let the exact reduced input be $\mathrm{v}_M^\star = V^{2^{-M}} - 1$, so that $\lg X = e + 2^M \lg(1 + \mathrm{v}_M^\star)$. Since $V \in (1, 2]$, we have

$$0 < \mathrm{v}_M^\star \le \rho_M, \qquad \rho_M := 2^{2^{-M}} - 1 \ \le 2^{-M}.$$

We approximate $\lg(1 + \mathrm{v}_M^\star)$ by the N-term Taylor polynomial in Horner form, $\mathcal{L}_N(\mathrm{v}) = \mathrm{v}(c'_1 + \mathrm{v}(c'_2 + \cdots + \mathrm{v}c'_N))$, with $c'_i = \frac{(-1)^{i+1}}{i \ln 2}$. The circuit output is $\widetilde{\lg(X)} = e + 2^M \widetilde{\mathcal{L}}_N(\mathrm{v}_M)$, where v_M is the implemented reduced input produced by the SQRT chain.

Let Δ_p denote the arithmetic quantization unit in $Q2.p$, i.e., $\Delta_p = 2^{-p}$ for truncation and $\Delta_p = 2^{-(p+1)}$ for round-to-nearest, ties-to-even. We also let $K \in \{2, 1\}$ denote the number of arithmetic quantizations per Horner level: $K = 2$ for separate multiply-add and $K = 1$ for FMA-like evaluation. Finally, let $\varepsilon_{\mathrm{sqrt}}$ be the local SQRT error bound, namely $|S_{i+1} - \sqrt{S_i}| \le \varepsilon_{\mathrm{sqrt}}$. For the present implementation, $\varepsilon_{\mathrm{sqrt}} = 2^{-p}$.

A.1 Error Decomposition and Bounds

The total forward error is decomposed as

$$\begin{aligned}\left|\widetilde{\lg(X)} - \lg X\right| &= \left|2^M(\widetilde{\mathcal{L}}_N(\mathrm{v}_M) - \lg(1 + \mathrm{v}_M^\star))\right| \\ &\le 2^M(E_{\mathrm{trunc}} + E_{\mathrm{arith}} + E_{\mathrm{coef}} + E_{\mathrm{rr}} + E_{\mathrm{round}}).\end{aligned}$$

The upper bounds for the individual components are derived below.

A.1.1 Taylor truncation. Let $R^{(\lg)}_{N+1}(\mathrm{v}) := \lg(1 + \mathrm{v}) - \mathcal{L}_N(\mathrm{v})$. Since $\lg(1 + \mathrm{v}) = \frac{1}{\ln 2} \sum_{k=1}^{\infty} (-1)^{k+1} \frac{\mathrm{v}^k}{k}$, the series is alternating with decreasing term magnitudes for $0 < \mathrm{v} \le \rho_M < 1$. Therefore,

$$E_{\mathrm{trunc}} = |R^{(\lg)}_{N+1}(\mathrm{v}_M^\star)| \le \frac{1}{\ln 2} \frac{(\mathrm{v}_M^\star)^{N+1}}{N+1} \le \frac{1}{\ln 2} \frac{\rho_M^{N+1}}{N+1}. \tag{5}$$

A.1.2 Arithmetic quantization along the Horner recurrence. For a fixed input v and the stored coefficients $\tilde{c}'_i$, define the exact Horner recurrence as $T_0(\mathrm{v}; \tilde{c}') = 0$ and $T_k(\mathrm{v}; \tilde{c}') = \mathrm{v}(T_{k-1}(\mathrm{v}; \tilde{c}') + \tilde{c}'_{N-k+1})$, so that, $T_N(\mathrm{v}; \tilde{c}') = \mathcal{L}_N(\mathrm{v}; \tilde{c}')$. Let $t_k(\mathrm{v})$ be the implemented recurrence, and $E_k(\mathrm{v}) := |t_k(\mathrm{v}) - T_k(\mathrm{v}; \tilde{c}')|$. In the non-FMA model, one Horner level incurs one multiplication quantization and one addition quantization, hence $E_{k+1}(\mathrm{v}) \le \mathrm{v}E_k(\mathrm{v}) + 2\Delta_p$. In the FMA-like model, one Horner level incurs only one quantization, hence $E_{k+1}(\mathrm{v}) \le \mathrm{v}E_k(\mathrm{v}) + \Delta_p$. Both cases are written as

$$E_{k+1}(\mathrm{v}) \le \mathrm{v}E_k(\mathrm{v}) + K\Delta_p, \qquad E_0(\mathrm{v}) = 0.$$

Unfolding the recurrence and substituting $\mathrm{v} = \mathrm{v}_M$ gives

$$E_{\mathrm{arith}} = E_N(\mathrm{v}_M) \le K\Delta_p \sum_{j=0}^{N-1} \mathrm{v}_M^j = K\Delta_p \frac{1 - \mathrm{v}_M^N}{1 - \mathrm{v}_M} \le K\Delta_p \frac{1 - \rho_M^N}{1 - \rho_M}. \tag{6}$$

A.1.3 Coefficient quantization. Let the stored coefficients be $\tilde{c}'_i = c'_i + \delta'_i$ with $|\delta'_i| \le \Delta_p$. Since $\mathcal{L}_N(\mathrm{v})$ is linear in $\{c'_i\}$, substituting $\mathrm{v} = \mathrm{v}_M$ gives

$$\begin{aligned}E_{\mathrm{coef}} = \left|\mathcal{L}_N(\mathrm{v}_M; \tilde{c}') - \mathcal{L}_N(\mathrm{v}_M; c')\right| &\le \sum_{k=1}^{N} \mathrm{v}_M^k |\delta'_k| \\ &\le \Delta_p \sum_{k=1}^{N} \mathrm{v}_M^k = \Delta_p \frac{\mathrm{v}_M(1 - \mathrm{v}_M^N)}{1 - \mathrm{v}_M} \le \Delta_p \frac{\rho_M(1 - \rho_M^N)}{1 - \rho_M}.\end{aligned} \tag{7}$$

A.1.4 Range-reduction-induced input error. Let $\eta_M := |\mathrm{v}_M - \mathrm{v}_M^\star|$. Since $\mathrm{v}_M = S_M - 1$ and $\mathrm{v}_M^\star = S_M^\star - 1$, we have $\eta_M = |S_M - S_M^\star|$. Define $r_i := |S_i - S_i^\star|$. Because the square-root map is $1/2$-Lipschitz on $(1, 2]$, the SQRT-chain error satisfies $r_{i+1} \le \frac{1}{2} r_i + \varepsilon_{\mathrm{sqrt}}$ with $r_0 = 0$. Unfolding the recurrence gives

$$\eta_M = r_M \le \varepsilon_{\mathrm{sqrt}} \sum_{j=0}^{M-1} 2^{-j} = 2\varepsilon_{\mathrm{sqrt}}(1 - 2^{-M}).$$

To propagate this perturbation through the Taylor polynomial, apply the mean-value theorem:

$$|\mathcal{L}_N(\mathrm{v}_M) - \mathcal{L}_N(\mathrm{v}_M^\star)| = |\mathcal{L}'_N(\xi_M)|\, \eta_M$$

for some $\xi_M \in (0, \rho_M]$. Moreover,

$$\mathcal{L}'_N(x) = \sum_{k=1}^{N} k\, c'_k x^{k-1}, \quad |\mathcal{L}'_N(x)| \le \frac{1}{\ln 2} \sum_{k=0}^{N-1} x^k = \frac{1}{\ln 2} \frac{1 - x^N}{1 - x}.$$

Hence

$$\begin{aligned}E_{\mathrm{rr}} = |\mathcal{L}_N(\mathrm{v}_M) - \mathcal{L}_N(\mathrm{v}_M^\star)| &\le \frac{1}{\ln 2} \eta_M \frac{1 - \rho_M^N}{1 - \rho_M} \\ &\le \frac{2}{\ln 2} \varepsilon_{\mathrm{sqrt}}(1 - 2^{-M}) \frac{1 - \rho_M^N}{1 - \rho_M}.\end{aligned} \tag{8}$$

A.1.5 Final rounding. The final rounding from $Q_{2.p}$ to $Q_{2.(F+M)}$ uses round-to-nearest, ties-to-even, so

$$E_{\mathrm{round}} \le 2^{-(F+M+1)}. \tag{9}$$

A.2 Master bound

Combining (5), (6), (7), (8), and (9), we obtain

$$\begin{aligned}\left|\widetilde{\lg(X)} - \lg X\right| \le 2^M \Bigg(&\frac{\rho_M^{N+1}}{(N+1)\ln 2} + \Delta_p (K + \rho_M) \frac{1 - \rho_M^N}{1 - \rho_M} \\ &+ \frac{2}{\ln 2} \varepsilon_{\mathrm{sqrt}}(1 - 2^{-M}) \frac{1 - \rho_M^N}{1 - \rho_M} + 2^{-(F+M+1)} \Bigg).\end{aligned} \tag{10}$$

For the present implementation, where the SQRT output, multiplier output, and coefficient storage are all truncated to p fractional bits, we set $\varepsilon_{\mathrm{sqrt}} = \Delta_p = 2^{-p}$. Using $\rho_M \le 2^{-M}$, we obtain the practical bound

$$\begin{aligned}\left|\widetilde{\lg(X)} - \lg X\right| \le &\frac{2^{-MN}}{(N+1)\ln 2} \\ &+ 2^{M-p} \left(K + 2^{-M} + \frac{2(1 - 2^{-M})}{\ln 2} \right) \frac{1 - 2^{-MN}}{1 - 2^{-M}} + 2^{-(F+1)}.\end{aligned} \tag{11}$$

Therefore, the Taylor truncation is mainly controlled by MN, while the fixed-point circuit error is mainly controlled by 2^{M-p}.